

run_pipeline.py > ...

```
1  import os
2  import pandas as pd
3  import matplotlib.pyplot as plt
4  import seaborn as sns
5  from datetime import datetime
6
7  # 1. SETUP PATHS & DIRECTORIES
8  RAW_DATA_PATH = "raw_data/raw_sales_data.csv"
9  CLEAN_DATA_DIR = "clean_data"
10 REPORTS_DIR = "reports"
11
12 # Ensure output directories exist
13 os.makedirs(CLEAN_DATA_DIR, exist_ok=True)
14 os.makedirs(REPORTS_DIR, exist_ok=True)
15
16 print("🚀 Starting Data Automation Pipeline...")
17
18 # =====
19 # STEP 1: DATA INGESTION & CLEANING
20 # =====
21 try:
22     # Load the messy data
23     df = pd.read_csv(RAW_DATA_PATH)
24     initial_rows = len(df)
25     print(f"📊 Successfully loaded {initial_rows} raw records.")
26
27     # A. Handle Missing Values
28     # Drop rows where critical identifiers (like Transaction ID or Customer ID) are missing
29     df = df.dropna(subset=['Transaction_ID'])
30
31     # Impute missing numeric values (e.g., Revenue) with the column median
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56 # Save cleaned data with a timestamped filename
57 timestamp = datetime.now().strftime("%Y%m%d")
58 clean_file_path = os.path.join(CLEAN_DATA_DIR, f"clean_sales_{timestamp}.csv")
59 df.to_csv(clean_file_path, index=False)
60 print(f"📁 Cleaned dataset saved to: {clean_file_path}")
61
62 except FileNotFoundError:
63     print(f"❌ Error: Could not find 'raw_sales_data.csv' in {RAW_DATA_PATH}. Please add your source file.")
64     exit()
65
66 # =====
67 # STEP 2: AUTOMATED REPORTING & VISUALIZATION
68 # =====
69 print("📊 Generating automated visual summaries...")
70
71 # Set visual style
72 sns.set_theme(style="whitegrid")
73 plt.figure(figsize=(10, 6))
74
75 # Visualization A: Revenue by Region (Bar Chart)
76 if 'Region' in df.columns and 'Revenue' in df.columns:
77     region_summary = df.groupby('Region')['Revenue'].sum().reset_index()
78
79     sns.barplot(data=region_summary, x='Region', y='Revenue', hue='Region', palette='Blues_r', legend=False)
80     plt.title('Automated Summary: Total Revenue by Region', fontsize=14, fontweight='bold')
81     plt.xlabel('Region', fontsize=12)
82     plt.ylabel('Total Revenue ($)', fontsize=12)
83     plt.tight_layout()
84
85 # Save chart
86 chart_path = os.path.join(REPORTS_DIR, f"revenue_by_region_{timestamp}.png")
87 plt.savefig(chart_path, dpi=300)

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# =====
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# =====
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83     plt.tight_layout()
84
85     # Save chart
86     chart_path = os.path.join(REPORTS_DIR, f"revenue_by_region_{timestamp}.png")
87     plt.savefig(chart_path, dpi=300)
88     plt.close()
89     print(f"📊 Chart successfully generated: {chart_path}")
90
91     # Visualization B: Executive Text Summary Report
92     report_path = os.path.join(REPORTS_DIR, f"executive_summary_{timestamp}.txt")
93     with open(report_path, "w") as report:
94         report.write("=====\n")
95         report.write(f" AUTOMATED EXECUTIVE PERFORMANCE REPORT - {datetime.now().strftime('%Y-%m-%d')}\n")
96         report.write("=====\n\n")
97         report.write(f"• Total Rows Processed Successfully: {cleaned_rows}\n")
98         if 'Revenue' in df.columns:
99             report.write(f"• Total Revenue Generated: ${df['Revenue'].sum():,.2f}\n")
100             report.write(f"• Average Transaction Value: ${df['Revenue'].mean():,.2f}\n")
101         if 'Transaction_ID' in df.columns:
102             report.write(f"• Unique Transactions Documented: {df['Transaction_ID'].nunique()}\n\n")
103
104         report.write("Top Performing Regions:\n")
105         if 'Region' in df.columns and 'Revenue' in df.columns:
106             top_regions = df.groupby('Region')['Revenue'].sum().sort_values(ascending=False)
107             for region, rev in top_regions.items():
108                 report.write(f" - {region}: ${rev:,.2f}\n")
109
110     print(f"📄 Text report successfully generated: {report_path}")
111     print("🏁 Pipeline executed successfully. Ready for next week's data!")

```